Memory Rewrites Do Not Drift, They Collapse: Detecting and Guarding Against Catastrophic Consolidation in Long-Term LLM Agent Memory

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Abstract

LLM agents often keep long-term memory as a text summary that an LLM rewrites after every session. The common worry is *drift*: small losses that compound with each rewrite. We test this on three long-term memory benchmarks, LoCoMo, Memora and LongMemEval, with two writer models from 30B to 550B parameters, and find no compounding. Repeatedly rewriting fixed content loses facts once and then reaches a fixed point, and streaming rewrites are not significantly worse than append-only notes on average. What does happen is *collapse*: a single rewrite discards at least half of the memory. On LoCoMo and Memora, 13 of 351 rewrites (3.7%) collapse, hitting 6 of 19 streams, and only 5 of the 13 collapses are ever followed by a return to the earlier length. Collapse depends on the writer and the data: a 550B writer from the same family never collapsed in 293 rewrites, while on LongMemEval's longer histories the 30B writer collapsed in 17.4% of rewrites. On held-out LoCoMo conversations, collapse drops accuracy from 23.3% (append-only notes) to 3.3%. Because a collapse appears as a sharp drop in length, it is cheap to detect. A guard that retries the rewrite once and keeps the longer attempt fires on 4.6% of rewrites, recovers 12 of 16 collapses, and keeps memory bounded. On a held-out replication whose rule was fixed in advance, it improves accuracy over plain streaming by 6.0 points (95% CI [+2.0, +11.0], p=0.002) and by 20.0 points on collapsed conversations. On LongMemEval, however, the guard did not help (17.4% vs. 4.3%, p=0.077), so it is a fix for rare collapses only. Adding retrieved raw turns to the guarded memory gave no significant further gain. Rewrite failures in agent memory are rare and abrupt rather than gradual, so they should be monitored and guarded, not averaged over.

1 Introduction

Long-term conversational agents cannot keep every past session in context, so many systems keep a consolidated memory that an LLM edits as new sessions arrive (Packer et al., 2023; Chhikara et al., 2025; Xu et al., 2025; Rasmussen et al., 2025). Each rewrite is lossy, so a natural hypothesis is that errors compound: facts erode a little with every pass until the memory no longer reflects the user. Recent work reports that consolidated memories can become faulty under continuous updates (Zhang et al., 2026) and that compaction loses exact-wording rules over rounds (Zerhoudi et al., 2026).

We set out to measure this drift curve directly and did not find it. We found a different failure mode with a different remedy. Most rewrites preserve or grow the memory; a few discard most of it in one step. These *collapses* are invisible in averages over many users, yet they cause the worst per-user outcomes, such as a memory that ends with "dogs' names: not explicitly listed" after earlier notes had named all three dogs.

Contributions.

- A controlled measurement of rewrite drift on LoCoMo (Maharana et al., 2024) and Memora (Uddin et al., 2026). Fixed-content rewriting loses facts once and then reaches a fixed point. Streaming rewrites are statistically indistinguishable from append-only notes on average (§5).
- The characterisation of *rewrite collapse*: rare single-step losses of at least half of the memory. We report rates, threshold sensitivity, recovery, a second writer model, and a third benchmark, LongMemEval (Wu et al., 2024) (§6).
- A length-triggered *collapse guard*, evaluated on a held-out replication with the rule fixed

before the run, plus a bounded variant that removes the memory-size confound. We also report where it fails: on LongMemEval, where collapses are frequent, it did not help (§7).

- Two practical analyses: whether adding retrieved raw turns to the guarded memory helps (§8), and the context and compute cost of each memory design (§9).

Code, prompts, per-question scores and analysis scripts are released.[1]

2 Related Work

Agent memory. MemGPT (Packer et al., 2023), Mem0 (Chhikara et al., 2025), A-Mem (Xu et al., 2025) and Zep (Rasmussen et al., 2025) maintain external memories that an LLM updates. REALM revisits stored memories when they are recalled and reports gains on LoCoMo and LongMemEval (Song et al., 2026). LoCoMo (Maharana et al., 2024), LongMemEval (Wu et al., 2024) and Memora (Uddin et al., 2026) evaluate recall, temporal reasoning and knowledge updates over many sessions; Memora also scores whether agents rely on outdated memories.

Degradation under rewriting. Zhang et al. (2026) show that memory utility first rises and then degrades as LLM consolidation proceeds, and they recommend gating consolidation. Zerhoudi et al. (2026) show that compaction keeps 53% of safety rules after one round and 10% after five, and they propose type-aware compaction. Colaco and Lahjouji (2026) frame compaction as a rate–distortion trade-off. Dongre et al. (2025) model multi-turn context drift as a bounded stochastic process and observe stable, noise-limited equilibria rather than runaway degradation. We see the same equilibrium for fixed content. Our distinct observation is that, in streaming conversational memory, the harmful losses are discrete events that a length signal detects, so a gate needs no classifier and no second model.

3 Problem Setup

A conversation arrives as sessions $s_1, \ldots, s_T$. A *streaming* memory starts empty, $M_0 = \emptyset$, and after each batch of sessions an LLM writer produces $M_t = W(M_{t-1}, s_t)$, a full rewrite that adds, updates, removes and merges facts under a nominal word budget. A reader answers questions from M_T alone. Let $|M|$ be the word count. We call step t a **collapse** when

$$|M_{t-1}| \geq L_{\min} \quad \text{and} \quad |M_t| \leq \rho\,|M_{t-1}|, \qquad (1)$$

with ρ=0.5 and $L_{\min}$=300 words, fixed before any guard experiment. A collapse *recovers* if some later $|M_{t'}| \geq |M_{t-1}|$. Note that length recovery does not mean the discarded facts come back; later sessions add new facts.

4 Experimental Setup

Data. *LoCoMo*: all 10 conversations (19–32 sessions each). Per conversation we sample 10 questions, stratified by category, whose evidence is at least 5 sessions old (adversarial category excluded). *Memora*: 9 personas from the weekly split, 10 questions each, scored with Memora's rubric checks (442 checks). Sessions are fed in batches of 10. *LongMemEval-S* (cleaned release): 30 instances, about 48 sessions each, stratified by type (multi-session 8, knowledge-update 8, temporal reasoning 8, single-session-user 6; abstention items excluded; seed 13). Sessions are fed in batches of 5, and assistant turns are truncated to 300 characters.

Memory arms. *k0*: append-only notes extracted from each session, never rewritten. k_n: the full k0 memory rewritten n times (fixed content). *stream*: after each batch, the writer rewrites the whole memory with a nominal 1,200-word cap. *Guards* (§7) and *raw/hybrid* retrieval arms (§8) are defined where they are used. All prompts are in Appendix A.

Models. Main writer and reader: Nemotron-3.5-Lightning-30B-A3B with reasoning disabled, temperature 0. Second writer: Nemotron-3-Ultra-550B-A55B, with the reader kept fixed so that differences come from the memory. Judge: gpt-oss-120b, scoring each answer against the gold answer (LoCoMo, LongMemEval) or each rubric check (Memora). Early pilots used gpt-oss-20b and Qwen3.8-27B as writers.

Statistics. Accuracy is the share of correct items. Confidence intervals and two-sided p-values come from a paired bootstrap that resamples questions (clustering a question's rubric checks), with 5,000 resamples (Efron and Tibshirani, 1994).

5 No Compounding Drift

Fixed content. In the LoCoMo pilot (30 questions, gpt-oss-20b writer), one rewrite cut memory

[1] https://github.com/selvin-paul-raj/M.E-Thesis (folder drift-memory).

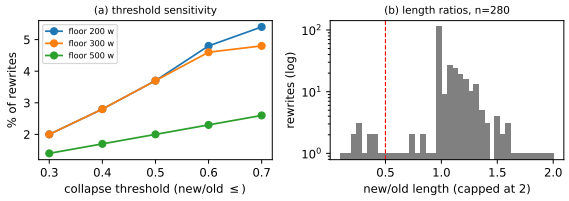


Figure 1: (a) Share of rewrites flagged as collapses for different thresholds ρ and length floors. (b) Distribution of new/old length over 280 rewrites that started from $\geq$300 words (log scale). The red line marks ρ=0.5.

from 1,129 to 684 words and kept 79% of k0 facts. Rounds 2–8 changed length and fact survival by at most 1–2%, and accuracy stayed within 30.0–36.7% for k=0 to 8. A small Memora pilot (2 personas, 20 questions, Qwen writer) fell from 58.3% at k=0 to 49.5% at k=2 and was identical at k=4 and k=8. The loss is front-loaded and then stops. Our pre-registered kill-switch for a compounding curve was not met in either pilot: both curves were flat after the first round.

Streaming. With Nemotron on all 9 Memora personas, k0, k_2 and stream scored 47.5, 48.2 and 50.9 (stream−k0 +3.4 [−2.8, +9.9], p=0.31). On held-out LoCoMo, stream averaged 18.0 against 26.0 for k0, and the whole gap came from three conversations (§6). Averages therefore show no steady erosion.

6 Rewrite Collapse

Rate. Over 19 Nemotron streams (9 Memora, 10 LoCoMo) and 351 rewrites, 13 rewrites collapsed (3.7%): 4 of 89 on Memora and 9 of 262 on LoCoMo. These hit 6 of 19 streams. Typical events shrank memories from 1,479 to 149 words or from 929 to 220.

Collapses are a separate mode, not a tail. Of the 280 rewrites that started from at least 300 words, 256 kept more than 90% of the length, 7 kept 70–90%, 4 kept 50–70%, and 13 kept at most 50% (Figure 1b). The distribution is bimodal: a large mass near 1.0 and a separate cluster at 0.1–0.4. Event counts move smoothly with the threshold (7 events at ρ=0.3, 19 at ρ=0.7 with a 200-word floor), and the set of affected streams barely changes (5–6 of 19; Figure 1a, Table 1). Our conclusions do not depend on the exact threshold.

Recovery. Only 5 of the 13 collapses were later followed by a memory at least as long as before. These were all on LoCoMo (conv-26 after 2 sessions, conv-42 after 12, and conv-47 three times

ρ	floor 200	floor 300	floor 500
0.3	7 (2.0)	7 (2.0)	5 (1.4)
0.4	10 (2.8)	10 (2.8)	6 (1.7)
0.5	13 (3.7)	**13 (3.7)**	7 (2.0)
0.6	17 (4.8)	16 (4.6)	8 (2.3)
0.7	19 (5.4)	17 (4.8)	9 (2.6)

Table 1: Collapse events (% of 351 rewrites) by threshold. Bold: the default used throughout.

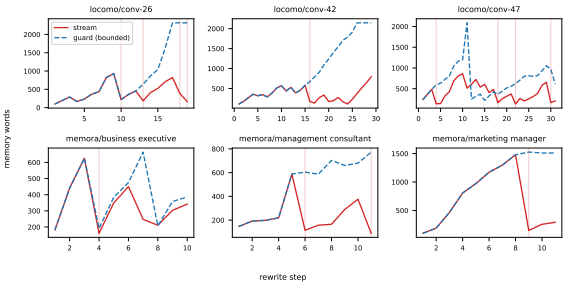


Figure 2: Memory length per rewrite for all 19 Nemotron streams; collapse events are marked.

after 4, 10 and 5 sessions). All 4 Memora collapses never recovered, and three streams ended at 89–342 words. Recovery is in length only. In conv-47 the stream collapsed four times; its final memory says the user's dogs' names are "not explicitly listed, but previously mentioned", while the k0 notes for the same conversation name Ned, Max and Daisy. Figure 2 shows the length trajectories.

Damage. On the three collapsed held-out LoCoMo conversations, stream accuracy was 3.3% against 23.3% for k0 (−20.0 [−36.7, −6.7], p=0.001). On the other seven, the arms were close. On Memora, the three collapsed personas averaged 41.9 for stream against 49.0 for k0, a gap hidden inside a non-significant mean.

Other writers and environments. The pilot writer gpt-oss-20b collapsed as well (4 events on one LoCoMo conversation). With the larger Nemotron-3-Ultra-550B-A55B writer and the same reader, *no* rewrite collapsed in 293 rewrites over 12 streams (all 10 LoCoMo conversations and 2 Memora personas; final memories 1,530–2,298 words). The guard therefore never fired, and guarded and plain streams scored the same (LoCoMo 39.0, Memora 85.0). On LoCoMo the larger writer's plain stream beat the 30B writer's by 21.0 points ([+11.0, +30.0], p<0.001) with the reader held fixed. Collapse is a property of the writer, and it fades with writer scale; it is not inherent to rewriting. On LongMemEval-S, with roughly 48 sessions per instance and the 30B writer, collapse was far

	k0	stream	g-app.	g-bnd.
Memora	47.5	50.9	56.8	55.2
collapsed (3)	49.0	41.9	58.7	54.2
LoCoMo, held-out	26.0	18.0	26.0	24.0
collapsed (3)	23.3	3.3	30.0	23.3
Pooled (190 q)	43.5	44.8	51.1	49.4

Table 2: Accuracy (%). g-app. and g-bnd. are the appending and bounded guards. Collapsed rows are the units whose plain stream collapsed.

Contrast	Subset	Δ [95% CI], p
g-app.−stream	Memora	+5.9 [+1.0, +11.6], .016
g-app.−stream	LoCoMo	+8.0 [+3.0, +14.0], <.001
g-bnd.−stream	Memora	+4.3 [−0.5, +9.9], .092
g-bnd.−stream	LoCoMo	+6.0 [+2.0, +11.0], .002
g-bnd.−stream	LoCoMo coll.	+20.0 [+6.7, +36.7], .002
g-bnd.−stream	Pooled	+4.6 [+0.6, +9.3], .020
g-bnd.−k0	Pooled	+5.9 [−0.8, +12.2], .086
g-bnd.−g-app.	Pooled	−1.7 [−3.3, −0.3], .023

Table 3: Question-clustered bootstrap contrasts (points).

more common: 39 of 224 rewrites (17.4%) in 21 of 23 completed instances, with only 11 of 39 regaining their earlier length. Longer, more topically scattered histories make collapse a frequent event rather than a rare one.

7 A Collapse Guard

A collapse has a simple signature, a large drop in length, so we guard against it when it happens. When a rewrite collapses, we re-issue it once with an explicit instruction to keep every existing fact unless the new session contradicts or retires it (Appendix A).

- **Guard-append**: if the retry also collapses, keep the old memory and append the new rewrite under "Recent updates".
- **Guard-bounded**: keep the longer of the two attempts and never append.

When no collapse occurs, both guards are identical to stream. Guard-append was designed after seeing the Memora collapses. We then fixed the rule and ran it on all 10 LoCoMo conversations as a held-out test. Guard-append can let memory grow without bound: on two LoCoMo conversations it reached 11,707 and 18,322 words, which confounds accuracy with memory size. Guard-bounded removes that confound. Its largest memory was 2,318 words, within the range of uncollapsed plain streams (up to 2,372).

Unit	k0	stream	g-app.	g-bnd.
LoCoMo conv-26	20.0	0.0	30.0	10.0
LoCoMo conv-42	30.0	10.0	40.0	40.0
LoCoMo conv-47	20.0	0.0	20.0	20.0
Memora bus. exec.	55.6	35.6	46.7	31.1
Memora mgmt. cons.	58.2	49.1	56.4	56.4
Memora mkt. mgr.	34.5	40.0	70.9	70.9

Table 4: Accuracy (%) on the six units whose plain stream collapsed.

Results. Tables 2 and 3 report accuracy and contrasts. On the held-out LoCoMo replication, both guards beat plain streaming, the bounded guard by +6.0 points (p=0.002). Almost all of the gain comes from the collapsed conversations (+20.0). The bounded guard matches k0 there (23.3 vs. 23.3). On uncollapsed units it is identical to stream by construction. A single retry recovered 12 of 16 bounded-guard triggers (75%), where recovery means keeping at least half of the old length. The bounded guard gives up 1.7 points against the appending guard. That gap is the price of keeping memory bounded, and the bounded guard is the variant we recommend. Pooled over both benchmarks, the bounded guard is the best bounded-size arm, but it does not significantly beat append-only notes (+5.9, p=0.086).

Per-unit view. Table 4 lists the six collapsed units. The bounded guard helps on five and hurts on one (Memora business_executive, 35.6 → 31.1), where the retry also collapsed and the longer attempt was still short. The appending guard handles that case better, which is the expected trade-off.

LongMemEval: the guard does not transfer. We ran the bounded guard unchanged on LongMemEval-S (23 of 30 planned instances completed; 7 runs were lost to provider errors). Here it fired on 48 of 224 rewrites (21.4%), and the retry kept at least half of the old length in 28 cases. Accuracy did *not* improve: stream 17.4%, guard 4.3% (−13.0 [−26.1, 0.0], p=0.077). The guarded memories were longer (mean 1,051 vs. 720 words) but answered fewer questions. With 23 questions the interval is wide, but the direction is clear, and we report this as a failure case: when collapses are frequent, forcing the writer to keep old content appears to crowd out the knowledge updates and temporal details LongMemEval asks about. The guard is a remedy for rare collapses, not for a writer that collapses on every fifth rewrite.

Subset	raw	hybrid	Δ [95% CI], p
LoCoMo (100 q)	31.0	39.0	+8.0 [−1.0, +17.0], .108
Memora (90 q)	54.3	50.9	−3.4 [−10.2, +3.5], .355
Collapsed units	53.5	51.4	−2.2 [−12.4, +7.5], .730
LongMemEval (23 q)	39.1	47.8	+8.7 [−8.7, +26.1], .444

Table 5: Raw retrieval alone vs. guarded memory plus raw retrieval (single-question reads).

8 Does Raw Evidence Help the Guarded Memory?

A natural extension is to pair the bounded-guard memory with retrieved original turns, so that facts lost in a rewrite can still be found. We test a *hybrid* reader context: the guarded memory plus the BM25 top-k raw turns for the question (k=10 on LoCoMo and LongMemEval, k=15 on Memora), against *raw*, the BM25 turns alone. These runs use single-question reads, whereas the earlier sections batch 10 questions per call, so we compare raw and hybrid only with each other.

Table 5 shows no significant gain from the combination. On LoCoMo the hybrid was 8 points ahead but the interval includes zero; on Memora it was 3.4 points behind. Per persona, the hybrid gained on four Memora personas and lost on three (largest loss: sales_manager, 44.9 → 22.4). A longer context with a consolidated summary on top does not reliably help this reader, so we do not claim the hybrid as an improvement. The negative result also bounds the guard's role: it repairs the memory, but it is not a substitute for retrieval when exact wording matters.

9 Deployment Cost

Table 6 gives the reader context each design produces. The collapse check is a word count and costs no model call. The bounded guard adds one writer call on each trigger: 16 of 351 rewrites (4.6%) on LoCoMo and Memora. The appending guard triggered on 40 rewrites (11.4%; 30 fell back to appending), and its memory grew to 18,322 words, so its reader cost is unbounded. The bounded guard's mean reader context (1,291 words) is close to plain streaming (1,070 on Memora, 1,533 on LoCoMo). On LongMemEval, where the guard fired on 21.4% of rewrites, the writer cost of the bounded guard rises accordingly, and the hybrid context is about three times the size of raw retrieval alone.

Env.	Arm	mean words	max words
LoCoMo	k0	1,217	2,098
LoCoMo	stream	1,533	2,372
LoCoMo	guard-append	4,634	18,322
Memora	k0	1,011	1,584
Memora	stream	1,070	1,556
Memora	guard-append	1,494	2,252
Both	guard-bounded	1,291	2,318
LongMemEval	stream	720	2,043
LongMemEval	guard-bounded	1,051	2,426
LongMemEval	raw (BM25)	500	689
LongMemEval	hybrid	1,561	2,847

Table 6: Reader context size per arm (words). The guard check itself needs no model call.

10 Discussion

The drift-as-erosion picture predicts a smooth decline that grows with the number of rewrites. Our data show something else: a stable equilibrium punctuated by rare catastrophic events. This has four practical consequences. First, average accuracy is a poor monitor. On Memora, three collapsed personas were hidden inside a non-significant mean. Systems should log memory length per rewrite and report the worst units, not only the mean. Second, collapses are cheap to detect. A word-count ratio needs no extra model call until it fires; the bounded guard re-runs 4.6% of rewrites and the appending guard 11.4%. Third, the remedy is local. A retry at the moment of collapse suffices, so the consolidation design does not need replacing. This complements Zhang et al. (2026), who argue for gating consolidation, by giving a concrete, inexpensive trigger. Fourth, the guard has limits. When the retry also collapses, a bounded guard cannot restore facts. On LongMemEval, where one rewrite in six collapsed, guarding made accuracy worse. The collapse rate itself is the useful monitor: near zero calls for nothing, a few percent calls for a guard, and a high rate calls for a stronger writer (a 550B writer removed collapse entirely in our runs).

11 Conclusion

LLM-rewritten agent memory does not erode steadily. It holds a fixed point and occasionally collapses. How often it collapses depends on the writer and the history: never in 293 rewrites with a 550B writer, 3.7% with a 30B writer on LoCoMo and Memora, and 17.4% on LongMemEval. When collapses are rare, a length-triggered retry with a bounded fallback repairs most of them and gave a significant held-out gain. When they are frequent,

the guard did not help, and a stronger writer is the better fix. Memory systems should log length per rewrite, guard against rare collapses, and treat a high collapse rate as a sign to change the writer.

Limitations

The guard was designed after inspecting the Memora collapses. Only the LoCoMo replication and the LongMemEval run are held out. The bounded variant was introduced after we saw the appending guard's length growth on LoCoMo, although its rule needs no tuning. Samples are small (90 + 100 + 23 questions; 7 planned LongMemEval instances and 7 Memora personas for the second writer were not completed because of provider errors), and collapses occur in few streams, so per-subset intervals are wide. The two main writers come from one model family (Nemotron), run at temperature 0 with one prompt. Other families, prompts and sampling temperatures are untested beyond two pilot writers. The judge is an LLM; strict grading together with frequent "unknown" answers lowers absolute LoCoMo accuracy. The 50%/300-word threshold is a heuristic. Writers exceed the nominal 1,200-word cap, so "bounded" means bounded by the writer's own behaviour. All benchmarks use synthetic or simulated dialogues; we did not test with real user logs. On LongMemEval we truncate assistant turns to 300 characters, which removes some answers to assistant-side questions.

Ethics Statement

We use only public benchmark data whose dialogues are synthetic or simulated. Memory systems that store personal facts raise privacy concerns. Collapse guards keep facts that would otherwise be dropped, so deletion requests must be honoured explicitly rather than left to lossy rewrites.

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A Prompts

Streaming update (writer). `You maintain a long-term memory for a chat assistant. Update the MEMORY with the NEW SESSION: add new facts (with absolute dates; resolve relative dates using the session date), update or remove facts that changed, merge duplicates, keep it organised and concise, one fact per line starting with '- '. Keep the whole memory under {cap} words. Output only`

the full updated memory. MEMORY: {mem} NEW SESSION: {sess}

Guard retry prefix. IMPORTANT: the current memory has {n} words. Your last attempt shrank it to {m} words and lost facts. Keep every existing fact unless the new session explicitly contradicts or retires it; merge new facts in. (followed by the streaming prompt)

k0 extraction. You maintain a long-term memory for a chat assistant. Below are new dialogue sessions (with dates). Write concise memory notes: one fact per line, starting with '- ', covering people, events (with absolute dates), preferences, plans, relationships. Resolve relative dates like 'yesterday' using the session date. Output only the notes.

Fixed-content rewrite ($\mathbf{k}_n$). Consolidate the memory below: merge duplicates, resolve conflicts, keep it organised and concise, one fact per line starting with '- '. Keep it under {cap} words. Output only the rewritten memory.

Reader. Answer each question using ONLY the context. Be short (a few words or a date). If the context does not contain the answer, write 'unknown'. Reply as JSON. The hybrid context is formatted as CONSOLIDATED MEMORY: ... RAW EVIDENCE (original messages retrieved for this question): ...

Judge. Grade each predicted answer against the gold answer. 1 if it conveys the same information (dates may differ in format), else 0. Reply as JSON.

B Reproducibility

All model calls are cached by prompt hash. Scripts: stream_pilot.py (streaming), a7_locomo.py (held-out LoCoMo), a8_retry_only.py (bounded guard), a9_analysis.py (thresholds, recovery, cost), a10_hybrid.py (raw/hybrid), a11_longmemeval.py (LongMemEval), a13_second_writer.py (second writer), boot.py (bootstrap). Per-question scores are released as CSV files.